

PrepBench ADF course: verification of the accuracy review, and corrections made

Audit date 2026-10-04 · Files changed: backend/app/content/packs/adf/v1.json , docs/research/adf-documentation-notes.md · Input reviewed: PrepBench_ADF_Accuracy_Audit_Report.pdf

A. Executive summary

The review was substantially correct on direction, and over-stated on two points. Of its ten numbered findings, **6 were confirmed** (ADF-001, 003, 004, 006, 009, 010), **3 were partly correct** (ADF-002, 005, 007), and **1 was not correct as stated** (ADF-008). Its two unnumbered remarks (Bicep coverage is thin; some numbers are brittle) were both confirmed. I found **9 further issues** the review missed.

- Every finding was checked against the Microsoft Learn page that owns the fact (34 pages, listed with URLs in section 30.2 of the research notes). The course was edited only where a page supported the change.
- The most important discovery is that **Microsoft's own pages disagree with each other** in six places (checkpoint keys, the self-hosted IR node limit, trusted service and the self-hosted IR, and others). Where they do, the course now teaches the safe common ground and says "check the current limit" instead of picking a side.
- No chapter, scenario or diagnostic question was added or removed, and no chapter N cross-reference changed, so the pack-shape tests and navigation are unaffected. New material went into existing chapters as labelled blocks.

B. Changes made

Source URLs are all under <https://learn.microsoft.com/en-us/azure/data-factory/> unless stated. "Notes §" points to docs/research/adf-documentation-notes.md.

Where	Original problem	Verified Microsoft behaviour	Change made	Source page	Severity
Copy activity chapter, "Connectors"	"Many databases can only be a source (Oracle, Netezza, Teradata, SAP)." Oracle is wrong; the repo's own notes already said Oracle can be a sink.	Oracle: Copy source and sink, Lookup, Script, on Azure and self-hosted IR. Netezza and Teradata are source-only in the published table. Oracle 1.0 is "at removal stage".	Replaced the list with "capabilities are per connector and version, look it up", keeping one dated illustrative contrast and a version-retirement note. Fixed the self-contradicting line in notes §25.	connector-oracle; connector-overview; copy-activity-overview	Critical
CDC chapter, "The rename trap"	Taught "renaming either resets it" as a universal rule.	Three statements: CDC concepts page (pipeline + activity name, rename resets); Data Flow activity page (new activities use a GUID, existing keep the old pattern; also "hash of data flow and pipeline names"). Own checkpoint key documented on both.	Rewritten to separate long-standing behaviour, newer behaviour and recommended practice; says the default is unsettled; recommends an explicit key and rehearsing a rename. Updated pitfalls, recovery table, diagnostic question and scenario 7 outcome.	concepts-change-data-capture; control-flow-execute-data-flow-activity	High

Integration runtimes chapter	Table said Azure IR "reaches on-premises data: No", self-hosted "Yes, without a VPN".	Standard Azure IR: public endpoints only (IP allowlisting is possible but "not a desirable solution" in high-security production). Azure IR with managed VNet: on-prem only via ExpressRoute or VPN plus a Private Link service and load balancer. SHIR: needs no extra settings, ExpressRoute or VPN, but its host must reach the source; the Azure leg is public HTTPS unless ExpressRoute/VPN plus Private Link are added.	Kept the beginner rule. Added a "simple rule and the nuance" block, a three-way comparison table, and rewrote the self-hosted section to state that it does not remove the need for a network path. Updated table cells, a scenario quiz explanation, and notes §6, §8, §16.	choose-the-right-integration-runtime-configuration ; concepts-integration-runtime; create-self-hosted-integration-runtime; connector-oracle	High
Integration runtimes and Security chapters	Managed VNet barely taught; terms never separated.	Three IR types per Microsoft; managed VNet is a creation-time setting of the Azure IR (cannot be converted). Managed private endpoints sit in that managed network and stay Pending until approved. A separate private endpoint to the factory service covers authoring and SHIR control traffic. Outbound to public endpoints is open.	"Not a fourth type" paragraph; four-term table (Azure IR, managed virtual network, managed private endpoint, private endpoint for the factory); exfiltration caveat.	managed-virtual-network-private-endpoint; data-factory-private-link	Medium
Sensitive data chapter; diagnostic question; pitfalls	"A customer-managed key can only be set on an empty factory."	Two routes: at creation in the portal (user-assigned managed identity), or after creation in the UI on an empty factory (not for managed-VNet factories). Once enabled it cannot be removed; key or version can change. Key Vault: same tenant and region, soft delete and purge protection.	Rewrote the "key decision too late" item, the controls table row, the diagnostic point and the pitfall row.	enable-customer-managed-key	High
CI/CD chapter; diagnostic question; pitfalls; scenario 4	"Only Dev in Git", "never edited by hand", "stop triggers", "publishing is all-or-nothing" all presented as product rules.	Git-only-on-Dev is a recommendation ("shouldn't"). Stopping triggers rests on a product behaviour: "Deployment can fail if you try to update active triggers". Selective publishing is unsupported "by design". Two supported ways to produce ARM templates: Publish button (<code>adf_publish</code>) or the <code>@microsoft/azure-data-factory-utilities</code> package (validate and export).	Rewrote both blocks with each point labelled Microsoft fact or recommended practice; added the automated route, Incremental vs Complete mode, global parameters, a Bicep pointer, and an interview takeaway. Reworded scenario 4 phrases that said the documentation "requires" stopping triggers.	continuous-integration-delivery; ...-improvement s; ...-automate-azure-pipelines; author-global-parameters; quickstart-create-data-factory-bicep	High

Performance and cost chapter; diagnostic question	"Four meters" list.	Microsoft lists three meters (orchestration runs, DIU-hours, vCore-hours) plus other costs (pipeline and external activity time, factory operations, SSIS IR). Per-pipeline billing has stated exceptions.	Replaced with a table: what costs money, what causes it, how to control it. Added per-pipeline billing limits and a caution on the duration table.	plan-manage-costs; azure.microsoft.com pricing page (meter names only)	Medium
Security chapter (new block)	Trusted service not covered.	Setting under Factory settings: Legacy or Modern. Modern recommended for new deployments. Legacy retires 1 August 2027. Five affected scenarios and three replacement access patterns listed. Modern's internals are not described.	New "Trusted service: Legacy and Modern" block with fact, what to switch to, what Microsoft does not say, recommended practice, takeaway; new pitfall row.	modern-trusted-service	High
Copy chapter, "Formats"	Listed Delta as a Copy format; omitted the Iceberg restriction.	Copy formats: Avro, Binary, delimited, Excel, Iceberg (ADLS Gen2 only), JSON, ORC, Parquet, XML. Delta is a data-flow format; Copy reaches Delta via the Databricks Delta Lake connector.	Corrected the list and added the explanation.	copy-activity-overview; connector-overview	Medium
Transform chapter	"Debug clusters take 5 to 7 minutes."	No such figure on the current pages. Debug TTL defaults to 60 minutes and is billed including TTL.	"Several minutes", plus the billed-idle-time fact.	concepts-data-flow-debug-mode; concepts-data-flow-overview	Low
Security chapter; Pipelines chapter; Triggers; Fabric; Fine-tuning	Smaller overstatements.	RBAC per entity is "not recommended", not impossible. ForEach cannot nest inside ForEach or Until. Custom event trigger exists. 120 activities is a default limit. Fabric offers a VNet data gateway alongside "managed VNet: to be determined".	Each corrected in place. Added "how much of this to memorise" block and a "how to read the labels" block (Microsoft fact, Recommended practice, Interview takeaway).	data-factory-private-link; control-flow-for-each-activity; concepts-pipeline-execution-triggers; fabric compare page	Low
Self-hosted IR node count (chapter, scenario 3 quiz and lenses)	"Up to four" stated as fixed.	Limits page: 4 (default and max). Fabric comparison page: "Up to 8 nodes (4 default)".	Now "Microsoft's limits page lists four; check the current limit"; quiz wording and explanation updated.	azure-subscription-service-limits; fabric compare page	Medium

Research notes (whole)	Notes contradicted the corrected course in §6, §8, §15, §17, §19, §23, §25, §27.	n/a	Every affected passage corrected and marked " (Corrected 2026-10-04)". Added §30: dispositions, 34 pages with URL, what each verified and its qualification, where Microsoft disagrees with itself, and what was not re-verified. Added a coverage-log row.	as above	Medium
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C. Review findings that were NOT correct as stated

- **ADF-008, "publishing is all-or-nothing is too absolute": not correct.** Microsoft's CI/CD page still lists selective publishing and cherry-picking as unsupported "by design", because entities depend on each other. Automated publish changes how the templates are generated (a build pipeline validates and exports them), not the granularity. The course's claim stands; what it lacked was the automated route, which is now added.
- **ADF-002, "outdated": overstated.** Microsoft's change data capture concepts page still teaches the rename rule. The honest finding is "too absolute and Microsoft is inconsistent", which is what the course now says.
- **ADF-007, "stop triggers is just policy": partly wrong.** Microsoft says the deployment can fail when it updates active triggers, so this is grounded in product behaviour, even though the script is a recommendation.
- **ADF-005, "Managed VNet can look like a fourth IR family": mis-aimed.** The course's three-type list matched Microsoft's. The real gap was that managed VNet was almost untaught. Fixed that instead.
- **ADF-001, version attribution:** the review ties Oracle sink support to connector 2.0; the page's capability table is not version-specific. The conclusion (Oracle is not source-only) is right.

D. New issues the review missed

- Delta listed as a Copy format; Iceberg is ADLS Gen2 only.
- "5 to 7 minutes" debug start-up is not on any current page.
- "Access can't be granted per pipeline" was stronger than Microsoft's "doesn't recommend".
- ForEach nesting limit also covers Until.
- Custom event trigger missing from the trigger table.
- "From any of about 100 connectors to any other" contradicted the course's own source-only point.
- 120-activity figure is a default limit.
- Self-hosted IR node limit is inconsistent between two Microsoft pages (4 vs 8).
- Fabric "managed VNet: to be determined" needed the VNet data gateway context.

E. Remaining risks (things that can go stale)

Area	Risk
Connector capabilities and versions	The Oracle/Teradata/Netezza contrast is dated October 2026. Oracle 1.0 retirement timeline is Microsoft's to change.
Trusted service	Retirement date (1 August 2027) comes from one page updated 2026-09-22. A Microsoft Q&A thread title suggests earlier confusion between 2026 and 2027; I saw only its title. The older data-access-strategies page conflicts about self-hosted IRs.
CDC checkpoint	Microsoft has not reconciled its three descriptions. Behaviour of GUID-keyed activities under rename is unverified.

Limits and defaults	120 activities, ForEach 20/50/100,000, Lookup 5,000 rows/4 MB, 256 DIUs, 45-day history, SHIR nodes. Flagged in the text as "check the current limit".
Pricing	Meter names only; no prices recorded. The pricing page was read through a summary, not as raw text.
Preview features	The CDC resource is still public preview. Fabric mount/upgrade paths and their GA dates were not re-fetched.
Not re-verified this audit	Fault-tolerance, consistency and type-conversion pages; migration worked examples; data-flow reconciliation functions; performance-tuning pages beyond DIU/parallel copy; Purview; white papers (full list in notes §30.5).
Scenarios 5 to 18	In this repo they are title and outcome stubs; their full text lives in the uncommitted prototype. I could not audit text I do not have. Only scenario 7's outcome line was reworded. They should be re-audited against this document before they are written into the pack.
UI behaviour	Portal paths (Factory settings, Manage hub) were taken from documentation, not observed in a live factory.

F. Validation

Check	Result
JSON parses	Yes. Written back with the original formatting (indent 2, no escaping, trailing newline).
Structure	21 chapters, 18 scenarios, 10 diagnostic questions, unchanged. Every block has only heading/md/table; every table row matches its header width; no bold markers in table cells (a repository rule). No TODO or placeholder text.
Cross-references	Every "chapter N" mention is exactly the set <code>test_content_packs.py</code> expects.
Backend tests	Full suite run from a throwaway git worktree (the learner's app is running, which would trip the real-database guard): 966 passed, 2 skipped, 0 failed (12 min 53 s). The two edited files were copied into the worktree, which was then removed. This includes <code>test_content_packs.py</code> (pack shape, cross-references, table cells) and the real-database isolation check.
Frontend unit tests	Full run: 1175 passed, 1 failed . The failure was <code>SystemDesignSetupPage.test.tsx</code> (a 20 s timeout, with three worker start-up timeouts in the same run), which ran while the backend suite was using the machine. Re-run alone it passed (6 of 6) . I am calling that a load/timing issue, as the repository guidance describes; I did not re-run the whole suite a second time, so the full run is not a clean single pass.
Not run	The Playwright browser suite (20 to 45 minutes; the user's app and dev server are running). Its guide test only asserts the 21-chapter count and chapter titles, which did not change.
Files changed	Two modified (<code>v1.json</code> , <code>adf-documentation-notes.md</code>) and this report added. The three untracked files that were already present (<code>PLAYWRIGHT-PERFORMANCE-REPORT.md</code> and two zips) were not touched. No application code changed. Nothing is committed; changes sit on branch <code>feat/lakehouse-lab-p1-3</code> .
Repo-wide sweep	Searched the whole repo for each corrected claim (Oracle source-only, rename/checkpoint, no VPN, empty factory, all-or-nothing, four meters, 5 to 7 minutes). Remaining hits are the corrections themselves.
Sources	Every changed Microsoft claim cites a page I fetched on 2026-10-04 (notes §30.2). Where I could not establish a fact, the text says so.

G. Final readiness rating

INTERVIEW-SAFE, with conditions. The written chapters no longer teach a claim I found to be false, and the dangerous absolutes (Oracle, on-premises reachability, "no VPN", CMK, CI/CD, cost, CDC checkpoint) are now qualified and labelled by claim type. It is **not** production-reference quality: several facts depend on volatile pages that disagree with each other, about a third of the supporting pages were not re-fetched, scenarios 5 to 18 are unaudited stubs here, and the figures need re-checking on a schedule (suggest each quarter, and immediately before August 2027 for the trusted-service date).